25900871 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCICH100 - FUNDAMENTALS OF INDUSTRIAL CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which of the following fuels is more likely to cause knocking in a gasoline engine? [A] / [CO2, CO4]

   a). Fuel with a high octane number b). Fuel with a low octane number

   c). Fuel with a high cetane number d). Fuel with high sulfur content

2. What is the main purpose of the isomerization process in petroleum refining? [A] / [CO2, CO4]

   a). To convert normal alkanes to branched alkanes b). To break down heavy oils into lighter fractions

   c). To remove impurities from crude oil d). To increase the viscosity of hydrocarbons

3. Which fraction from crude oil is the lightest? [K] / [CO2, CO4]

   a). Fuel oil b). Diesel

c). Gasoline d). Bitumen

4. Which of the following is the primary element that distinguishes steel from cast iron? [U] / [CO3]

a). Carbon b). Silicon

c). Chromiun d). Manganese

5. Rusting of iron is an example of: [U] / [CO1]

a). Acid-base reaction b). Oxidation-reduction reaction

c). Displacement reaction d). Precipitation reaction

6. Which process is used to remove gangue from an ore? [E] / [CO3]

a). Calcination b). Froth flotation

c). Smelting d). Electrolysis

7. What is the primary component of clay minerals? [U] / [CO3]

a). Magnesium Silicates b). Hydrous Aluminum Silicates

c). Calcium Carbonates d). Potassium Feldspar

8. What is a common characteristic of self-assembled monolayers (SAMs)? [U] / [CO5]

a). They require high temperatures for formation. b). They are used to modify surface properties.

c). They cannot be formed on solid surfaces. d). They are always thick and porous.

9. Which term refers to the process where nanoparticles spontaneously organize into a structured form? [U] / [CO5]

a). Self-replication b). Self-assembly

c). Nanofabrication d). Aggregation

10. Which of the following is a characteristic of nanorods? [U] / [CO2]

a). Length greater than width b). Same length and width

c). Circular in shape d). No defined structure

[10x1 = 10]

Part B

Short Answer Type Questions

Answer any **4** Questions

Each Question carries **3** marks

11.	What are unit operations in a chemical process?	[U] / [CO1]
12.	Explain why safety measures are important in a chemical manufacturing plant.	[U] / [CO1, CO2]
13.	How does alloying help in reducing corrosion?	[U] / [CO1]
14.	What is the Bayer process, and why is it important in the production of alumina?	[U] / [CO3]
15.	Write any two applications of fullerenes.	[A] / [CO2]
16.	Compare the sizes of any three nanomaterials and classify them.	[U] / [CO5]

[4x3 = 12]

Part C

Short Essay Type Questions

Answer any **4** Questions

Each Question carries **7** marks

17.	Discuss the significance of purity in fine chemicals and the industries where fine chemicals are typically used.	[U] / [CO1]
18.	Evaluate the impact of process parameters (temperature, pressure, catalyst) on the rate and selectivity of a chemical reaction.	[E] / [CO1, CO2]
19.	Describe the electrochemical process of rusting in detail. Explain the reactions that occur at the anode and cathode during rust formation and the conditions necessary for this process.	[U] / [CO1]
20.	Discuss the significance of regenerated cellulose (viscose) in the textile industry.	[U] / [CO4]
21.	Explain the underlying principles of chemical vapor deposition (CVD) and describe its process in detail.	[U] / [CO5]
22.	What are the limitations of the 'top-down' approach in nanomaterial synthesis? Provide examples of industries where this method is commonly used.	[U] / [CO5]

[4x7 = 28]